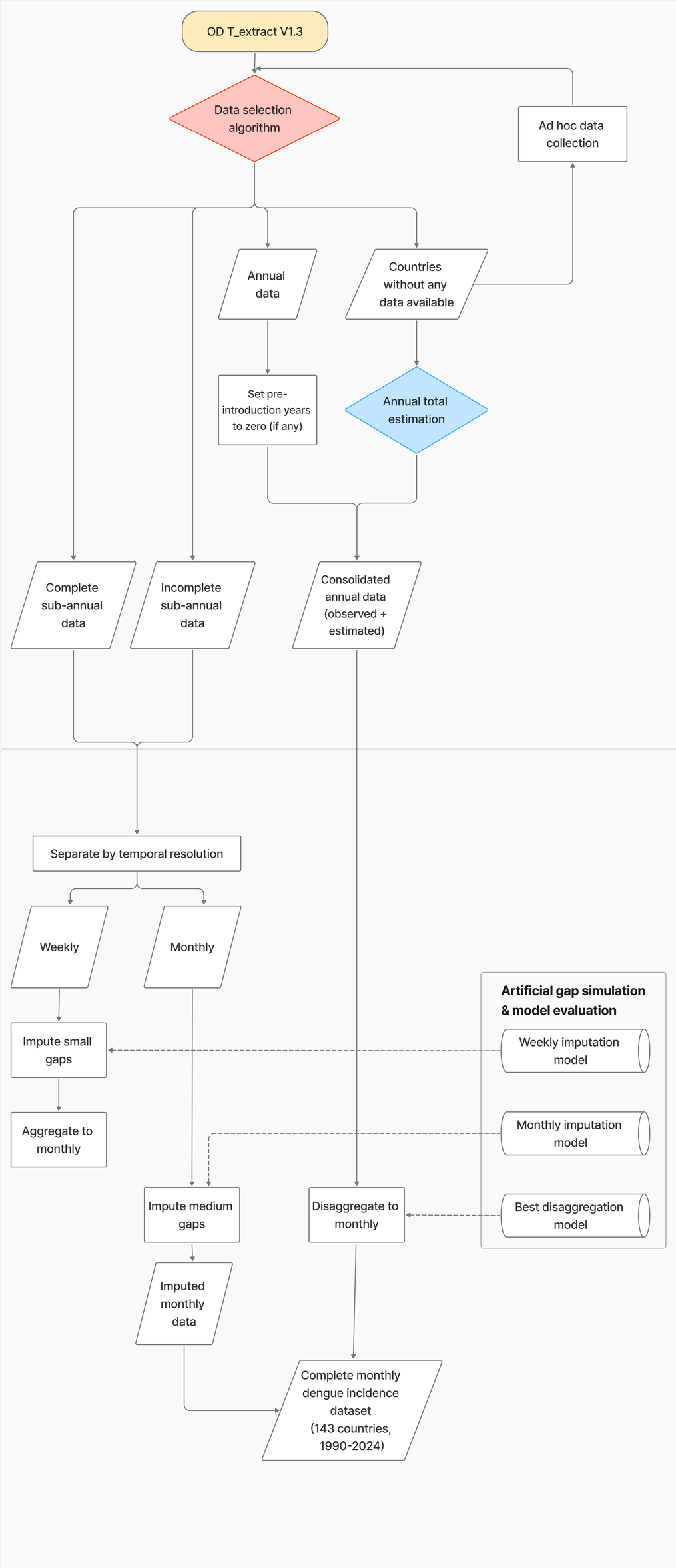
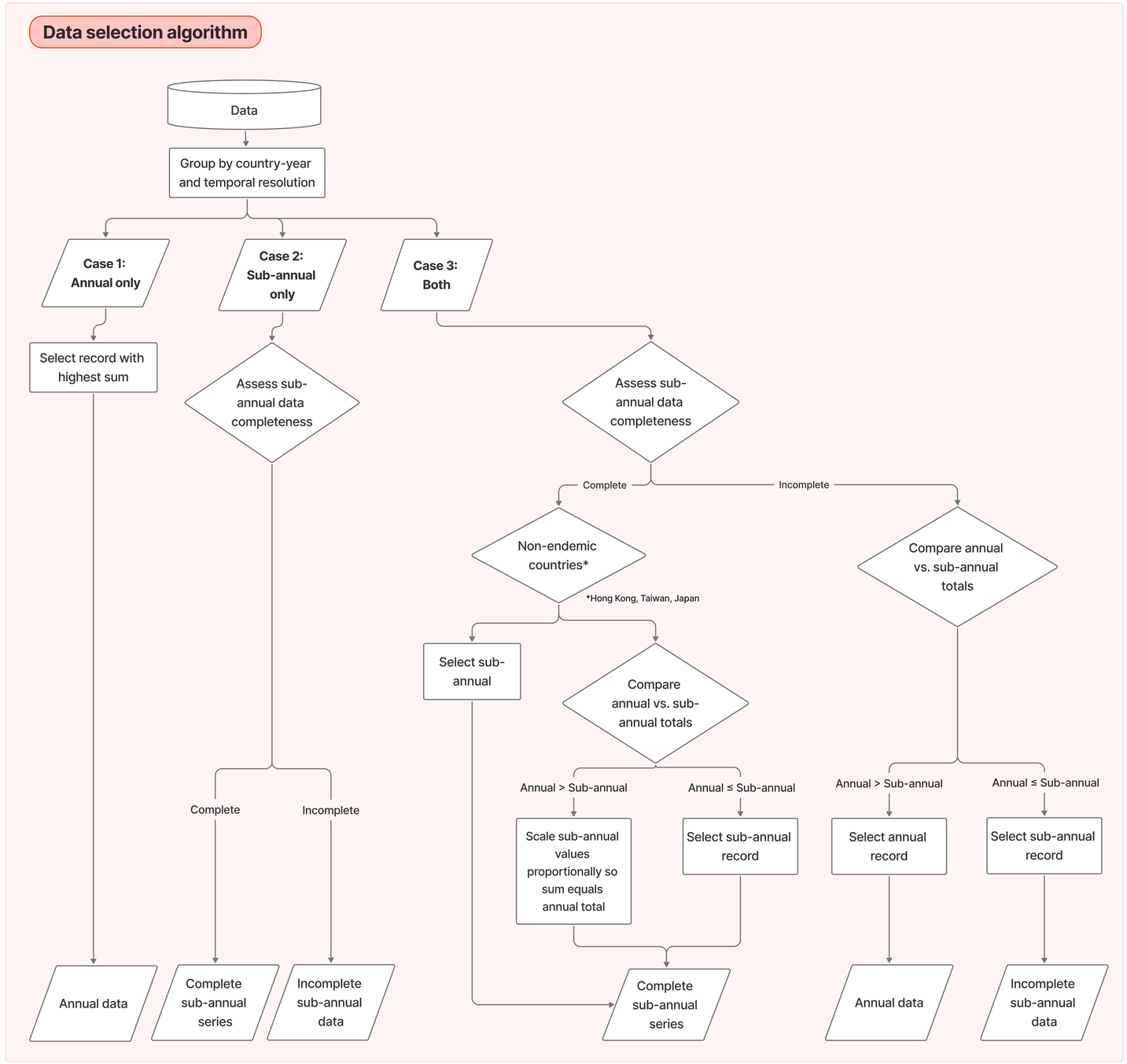
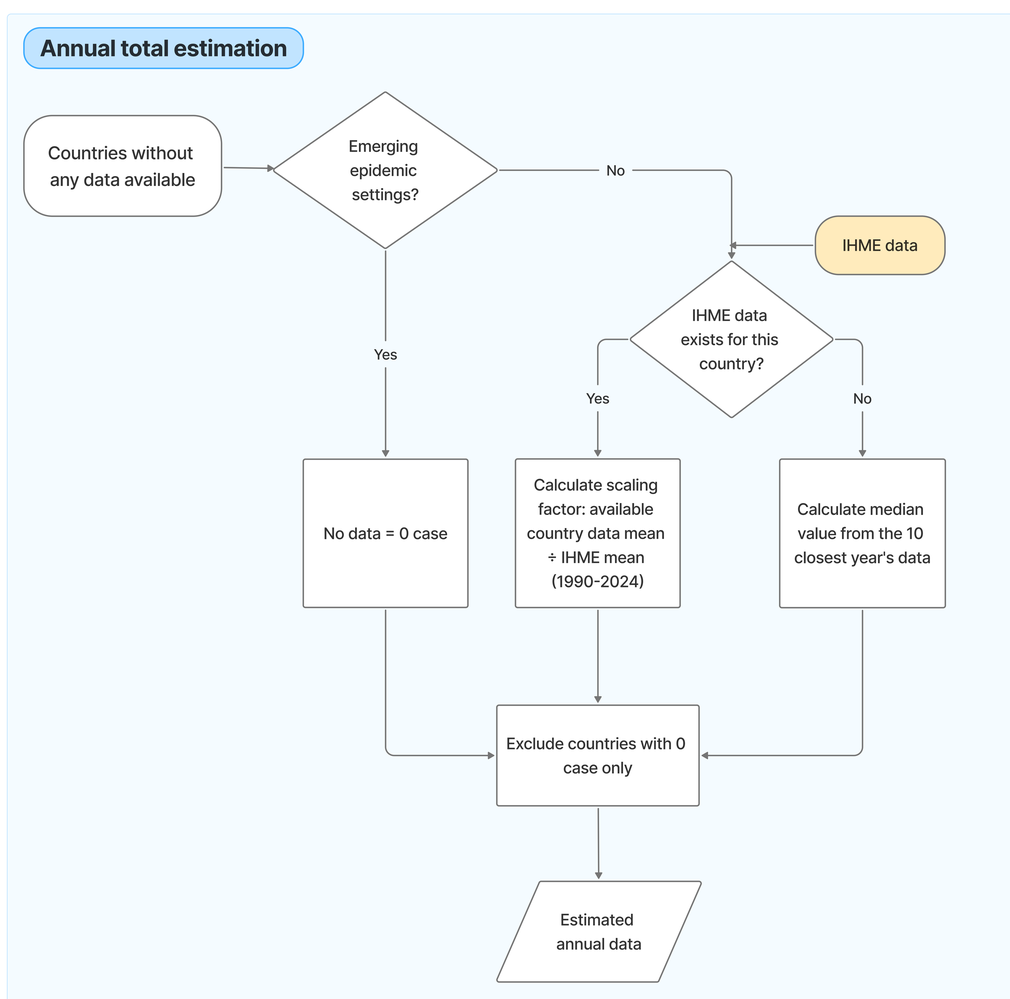

OD T_extract V1.3
Data selection algorithm
Ad hoc data collection
Annual data
Countries without any data available
Set pre-introduction years to zero (if any)
Annual total estimation
Complete sub-annual data
Incomplete sub-annual data
Consolidated annual data (observed + estimated)
Separate by temporal resolution
Weekly
Monthly
Impute small gaps
Aggregate to monthly
Impute medium gaps
Disaggregate to monthly
Imputed monthly data
Complete monthly dengue incidence dataset (143 countries, 1990-2024)
Artificial gap simulation & model evaluation
Weekly imputation model
Monthly imputation model
Best disaggregation model


Data selection algorithm
Data
Group by country-year and temporal resolution
Case 1: Annual only
Case 2: Sub-annual only
Case 3: Both
Select record with highest sum
Assess sub-annual data completeness
Assess sub-annual data completeness
Complete
Incomplete
Non-endemic countries*
*Hong Kong, Taiwan, Japan
Compare annual vs. sub-annual totals
Select sub-annual
Compare annual vs. sub-annual totals
Annual > Sub-annual
Annual ≤ Sub-annual
Annual > Sub-annual
Annual ≤ Sub-annual
Complete
Incomplete
Scale sub-annual values proportionally so sum equals annual total
Select sub-annual record
Select annual record
Select sub-annual record
Annual data
Complete sub-annual series
Incomplete sub-annual data
Complete sub-annual series
Annual data
Incomplete sub-annual data


Annual total estimation
Countries without any data available
Emerging epidemic settings?
No
IHME data
Yes
IHME data exists for this country?
Yes
No
No data = 0 case
Calculate scaling factor: available country data mean ÷ IHME mean (1990-2024)
Calculate median value from the 10 closest year's data
Exclude countries with 0 case only
Estimated annual data